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Parametric Design of Easy-Connect Pipe Fitting Components Using Open-Source CAD and Fabrication Using 3D Printing

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Abstract: The amount of non-revenue water, mostly due to leakage, is around 126 billion cubic meters annually worldwide. A more efficient wastewater management strategy would use a parametric design for on-demand, customized pipe fittings, following the principles of distributed manufacturing. To fulfill this need, this study introduces an open-source parametric design of a 3D-printable easy-connect pipe fitting that offers compatibility with different dimensions and materials of pipes available on the market. Custom pipe fittings were 3D printed using a RepRap-class fused filament 3D printer, with polylactic acid (PLA), polyethylene terephthalate glycol (PETG), acrylonitrile styrene acrylate (ASA), and thermoplastic elastomer (TPE) as filament feedstocks for validation. The 3D-printed connectors underwent hydrostatic water pressure tests to ensure that they met the standards for residential, agricultural, and renewable energy production applications. All the printed parts passed numerous hydrostatic pressure tests. PETG couplings can tolerate up to 4.551 ± 0.138 MPa of hydrostatic pressure, which is eight times greater than the highest standard water pressure in the residential sector. Based on the economic analysis, the cost of 3D printing a pipe coupling is from three to seventeen times lower than purchasing a commercially available pipe fitting of a similar size. The new open-source couplings demonstrate particular potential for use in developing countries and remote areas.



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1. Introduction

Due to the growing world population [1,2] and water shortages caused by the climate change crisis [3], there is an increasing demand for drinking water. Global water challenges are also compounded by a staggering amount of water loss from leaks. The amount of non-revenue water (NRW) is around 126 billion cubic meters annually worldwide, which is almost 3.4 times larger than the total annual freshwater withdrawal in Canada [4]. Leakages resulting in treated water turning into wastewater are considered non-revenue water and represent serious challenges for water management systems. Having leaks not only wastes valuable treated water, but also places a burden on water supply infrastructures to provide extra pressure to prevent any water contamination [5]. Based on previous studies [6,7], it is well established that the piping system in the majority of developing countries is old and suffers from degradation due to corrosion and environmental conditions [8].

These conditions lead to considerable leakages in the supply system, which results in 346 million cubic meters of global non-revenue water per day [9]. Due to the high cost of standard piping equipment and fittings relative to the incomes in developing countries, old components are often not replaced, or non-standard approaches are employed for repair. These choices can waste a significant amount of drinking water and lead to numerous diseases due to water contamination [7,10]. Furthermore, pipe fittings are costly, ranging from CAD 4.80 [11] for lower water pressure, such as domestic applications, to around CAD 18.50 [12] for high pressures and temperatures in industrial applications, according to the specific type of connection and their supply chain models that are often based on traditional centralized manufacturing [13–17]. In addition, due to expensive maintenance services in developed countries, minor leaks are often neglected, which results in both wasting natural resources and higher pumping costs [18].

The Sustainable Development Goals (SDGs) aiming to protect the Earth and human lives and end poverty for everyone all over the globe were established in 2015, with 17 goals to achieve by 2030 [19]. Shortcomings in achieving these goals have become more apparent as almost two-thirds of the time to complete them has passed. This fact is especially true for goal 6 (SDG6), “Ensure access to water and sanitation for all”, which focuses on the sustainable management of water resources and sanitation worldwide. Despite all the efforts, approximately 30% of the global population has remained without access to reliably treated drinking water, and around half of humanity faces water scarcity for at least 1 month every year [20,21]. Financial challenges such as underfunding, funding inequality, and financial sustainability play a major role in proper water management [22]. Considering that the global financial support needed to achieve SDG6 is estimated to be as high as USD 1.1 trillion per year [23] and the fact that more than 2 million Americans, Canadians, and Australians are also facing a lack of access due to a lack of indoor plumbing or inadequate wastewater disposal [24], the importance of introducing low-cost and sustainable solutions to address water management challenges and to reduce the amount of NRW globally is evident.

The centuries-long trend in centralized manufacturing has been reversed [25] by the introduction of low-cost distributed and localized [26] additive manufacturing (AM). Over the last forty years, AM has reached different industrial sectors thanks to the available materials, ranging from thermoplastics to ceramics and metal alloys, and processes. Following the ASTM standard, AM processes can be divided into seven main categories: binder jetting (BJT), directed energy deposition (DED), material extrusion (MEX), material jetting (MJT), powder bed fusion (PBF), sheet lamination (SHL), and VAT photopolymerization (VPP) [27]. In this regard, the self-replicating rapid prototype (RepRap) project [28–30] galvanized the entire AM industry thanks to its open-source approach [31], highlighting the potential of AM, especially MEX fused filament fabrication (FFF), to expand non-centralized manufacturing models. MEX printing can be complex [32], but distributed AM [33] has been shown to be a lower-cost solution than traditional manufacturing across a wide range of applications [34]. Distributed MEX AM applications include spare parts [35] and consumer goods [36] or high-end scientific [37,38] and medical equipment [39,40]. AM is particularly adept at solving distributed manufacturing problems using customized parts [41–43], making on-demand customization affordable for small production batches, e.g., repairs. Thus, introducing 3D-printed plastic pipe fittings could be a potential application for MEX FFF AM in water management. As open-source sharing of 3D-printable design files has been shown to facilitate sustainable development [44,45], the technical and economic problems associated with leakage management in water supply systems indicate a promising potential for open-source approaches to mitigate the cost of repair and maintenance in developed countries and standard equipment shortages in remote locations.

Despite this potential application, only two works have explored the capability of using MEX FFF AM to produce piping and fluidic components. In 2018, a study investigated the technical and economic feasibility of EcoPrinting technology by recycling waste polymers and turning them into 3D printing filaments for humanitarian aid components to prevent water leakage in a village disconnected from the grid located in a remote jungle area [46]. According to best practices for sustainable distributed AM [47–49], all the equipment used in this study was solar-powered [50], and waste plastic converted via single-screw extruders (recyclebots) [51,52] was used to produce 3D printing feedstock to tighten the circular economy loop [53]. During the investigations, designs were modified to reach the goal of having a near-zero carbon footprint [46], and 12 leakages were found and replaced by the 3D-printed pipe couplings designed by the authors. The leakages were fully addressed, although the testing process and the amount of pressure tolerated by the 3D-printed components were not reported. Three years later, Price et al. [54] conducted an experimental study that involved designing and 3D printing an open-source pipe connector for the flow chemistry field. The design consisted of a male nut and four connectors with different configurations 3D printed in polylactic acid (PLA), as well as a wetted 3D-printed part in polyether ether ketone (PEEK) to prevent any reactions with working fluid. A ferrule was also 3D printed using polypropylene (PP) filaments to prevent leakage. Although the design can perform highly effectively under high pressures, it may not be applicable to water supply systems in residential or industrial sectors due to the difference between the size of the pipe used in the study and the standard pipe diameters for those applications [54].

Although results in the previous literature are promising, further research is needed to implement 3D-printed fittings in real applications. A more efficient wastewater management strategy would be to introduce an open-source parametric design for on-demand customized pipe fittings, following the principles of distributed manufacturing. This solution would be economically feasible, easy to connect, and compatible with pipes made of different materials and of dimensions so that individuals with any skillset could repair their water supply system. Furthermore, these designs need to be validated through testing.

To fulfill this need, this study introduces an open-source parametric design of a 3D-printable easy-connect pipe fitting that offers compatibility with different dimensions and materials of pipes available on the market. Different pipe fitting 3D models were generated by using the new parametric design. The custom pipe fittings were then 3D printed using a RepRap-class MEX FFF 3D printer, selecting PLA, polyethylene terephthalate glycol (PETG), acrylonitrile styrene acrylate (ASA), and thermoplastic elastomer (TPE) as filament feedstocks for the validation. The 3D-printed connectors underwent hydrostatic water pressure tests to ensure that they met the standards for residential, agricultural, and renewable energy production applications. The manufacturing times and costs were then quantified, and an economic analysis of the new open-source couplings was performed to demonstrate their potential use in developing countries and remote areas. The results are discussed in the context of functional distributed manufacturing.

2. Materials and Methods

2.1. Open-Source Parametric Design

The design and implementation of the pipe fitting model program was written in OpenSCAD 2021.01 [55] and is organized into distinct modules and functions across several files, each with a specific role in constructing the final fitting.

Parametric design, as applied here, serves as a powerful paradigm in engineering as it enables flexible, adaptive workflows by treating the geometric properties of designs as variables interconnected in a network of dependencies. This approach ensures that modifi-

cations to one parameter propagate across the model, facilitating rapid customization and iteration without redundant drafting efforts. The power of this method extends beyond the geometric modeling implementation, as it provides a strategic framework for integrating design, performance, and fabrication [56]. The program reflects this through its incorporation of modular objects and a parameterized workflow to ensure that designs maintain consistency while adapting to user-defined input. Parametric malleability, central to this methodology, supports mass customization and optimization, where the underlying design schema (genotype) can generate numerous context-specific variations (phenotypes) [57]. For a detailed explanation of the use of OpenSCAD to perform parametric design of the pipe fitting see Appendix A.

A graphical overview of the presented workflow for parametric pipe fittings (OpenSCAD definition) can be found in Figure 1, providing an overview of the organization, logic, and functionality of the computational engineering model used to obtain the customized mesh 3D models of the pipe fittings. The open-source parametric design of this study is available on the Open Science Framework (OSF) under GNU General Public License 3.0 [58]. The approach used ensured that as the components were tightened down, they were sealed, and there were no sacrificial components.

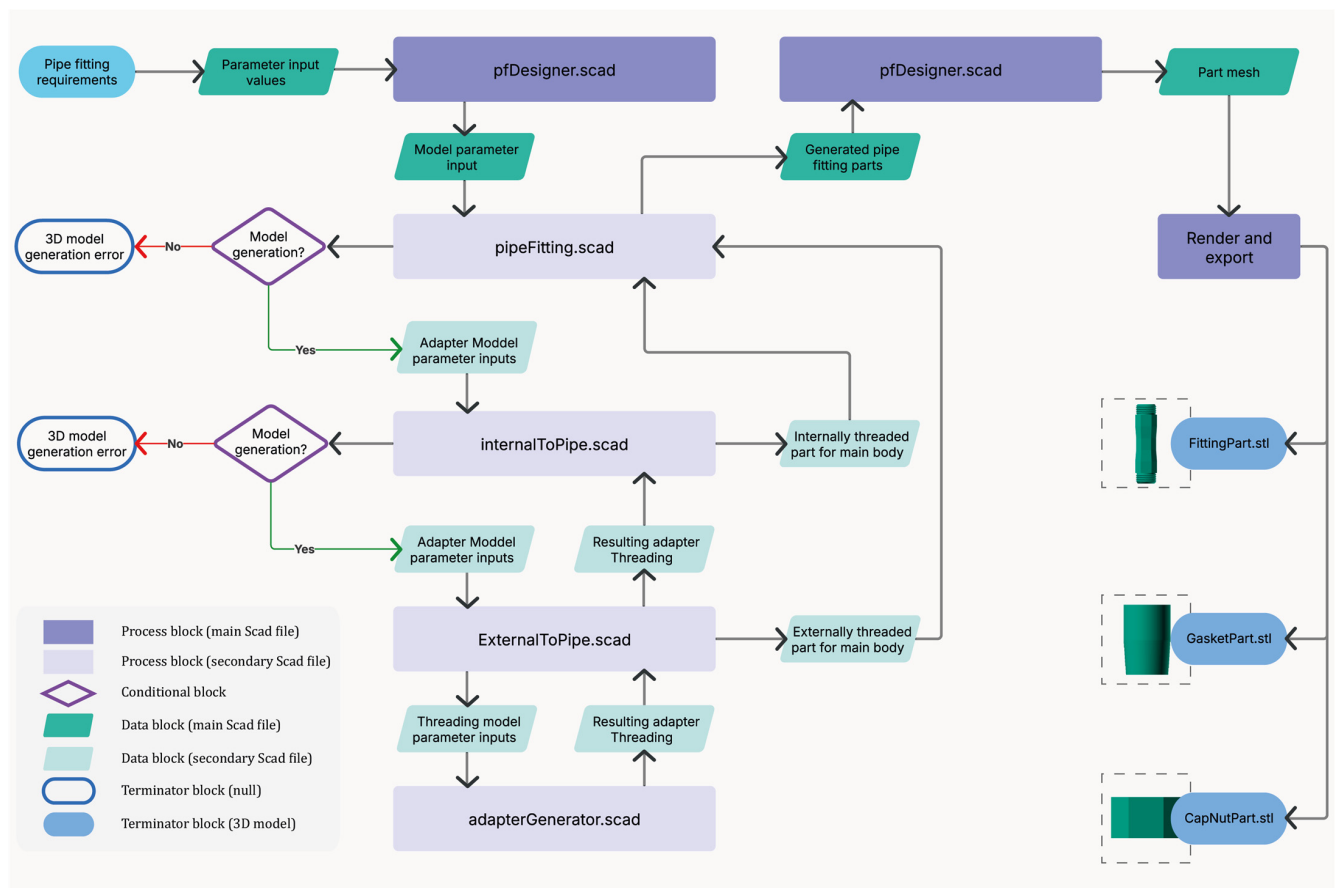


Figure 1. Pipe fitting program workflow (OpenSCAD definition) for the customization of the mesh 3D models (STL files).

The parametric design used *threadlib* [59] to generate standard thread types. A summary of the available thread types for the pipe fittings can be found in Table 1. In addition to the built-in thread types listed below, the library offers a method to extend it via additional thread types according to the desired thread standard. The customization interface can be accessed through the '*pfDesigner.scad*' file in the OSF repository [58], which

contains high-level input parameters accessible to the user and is connected to specific OpenSCAD definitions for the customization of specific features, e.g., “*pipefitting.scad*”, “*internalToPipe.scad*”, “*externalToPipe.scad*”, and “*adapterGenerator.scad*”. These parameters are fully summarized in Table 2, whereas Figure 2a shows the connection between each input parameter and the corresponding feature or part of the 3D model, where the dots indicate the specific areas affected by each parameter change. Further details are available in Appendix A (Figure A1).

Table 1. Thread standards available to the user during the customization.

Thread Type	Range	Details
Metric Threads	M0.25 to M600	Coarse, fine, and super-fine pitches
Unified Inch Screw Threads (UNC, UNE, UNEF)	Various (UNC, UNE, UNEF) including 4-UN, 6-UN, 8-UN, 12-UN, 16-UN, 20-UN, 28-UN, and 32-UN	All threads are class 2 threads
BSP Parallel Thread (G)	G1/16 to G6	All threads are class A threads
PCO-1881 and PCO-1810	Specific to PET-bottle threads	Not applicable
Royal Microscopical Society’s Thread (RMS)	Typical RMS	Not applicable

Table 2. User-accessible input parameters/variables (primary variables define the design, secondary variables modify it, derived variables result from the primary variable combination, and utility variables aid in exporting the final design).

N.	Input Name	Range	Type	Description
1	<i>input_dia</i>		Primary	The diameter of the input pipe being fit.
2	<i>gasket_thickness_upper</i>		Primary	Upper bound on desired gasket thickness.
3	<i>gasket_thickness_lower</i>		Primary	Lower bound on desired gasket thickness.
4	<i>gasket_extra_length</i>		Secondary	Additional length added to the end of the gasket such that the cap nut compresses it when assembled.
5	<i>tol_pipe</i>		Secondary	Tolerance between the pipe and the fitting.
6	<i>tol_gasket</i>		Secondary	Tolerance between the pipe and the gasket.
7	<i>turns</i>		Secondary	Number of turns for the adapter threading.
8	<i>wall_thickness</i>		Secondary	Thickness of the fitting walls.
9	<i>lower_length</i>		Secondary	Length of the middle pipe of the fitting.
10	<i>mid_height</i>		Secondary	Length of the transition from the middle pipe to the adapter threading begins.
11	<i>cap_thickness</i>		Secondary	Thickness of the cover part of the cap nut.
12	<i>nut_wall_thickness</i>		Secondary	Thickness of the cap nut walls.
13	<i>entry_chamfer</i>	True or false	Secondary	Chamfer around the entry lip of the adapter threads.
14	<i>style</i>	String input	Secondary	Available options: “Hexagon”, “Cone”, or “Circular”.
15	<i>selectedThread</i>	Integer	Derived	The desired adapter thread the user wishes to export, must be in the range of possibilities.
16	<i>export</i>	True or false	Utility	Set to true to remove all design possibilities and only display the selected one.
17	<i>selectedPart</i>	String input	Utility	Available options: “fitting”, “nut”, “gasket”, or “all” to select objects for export.

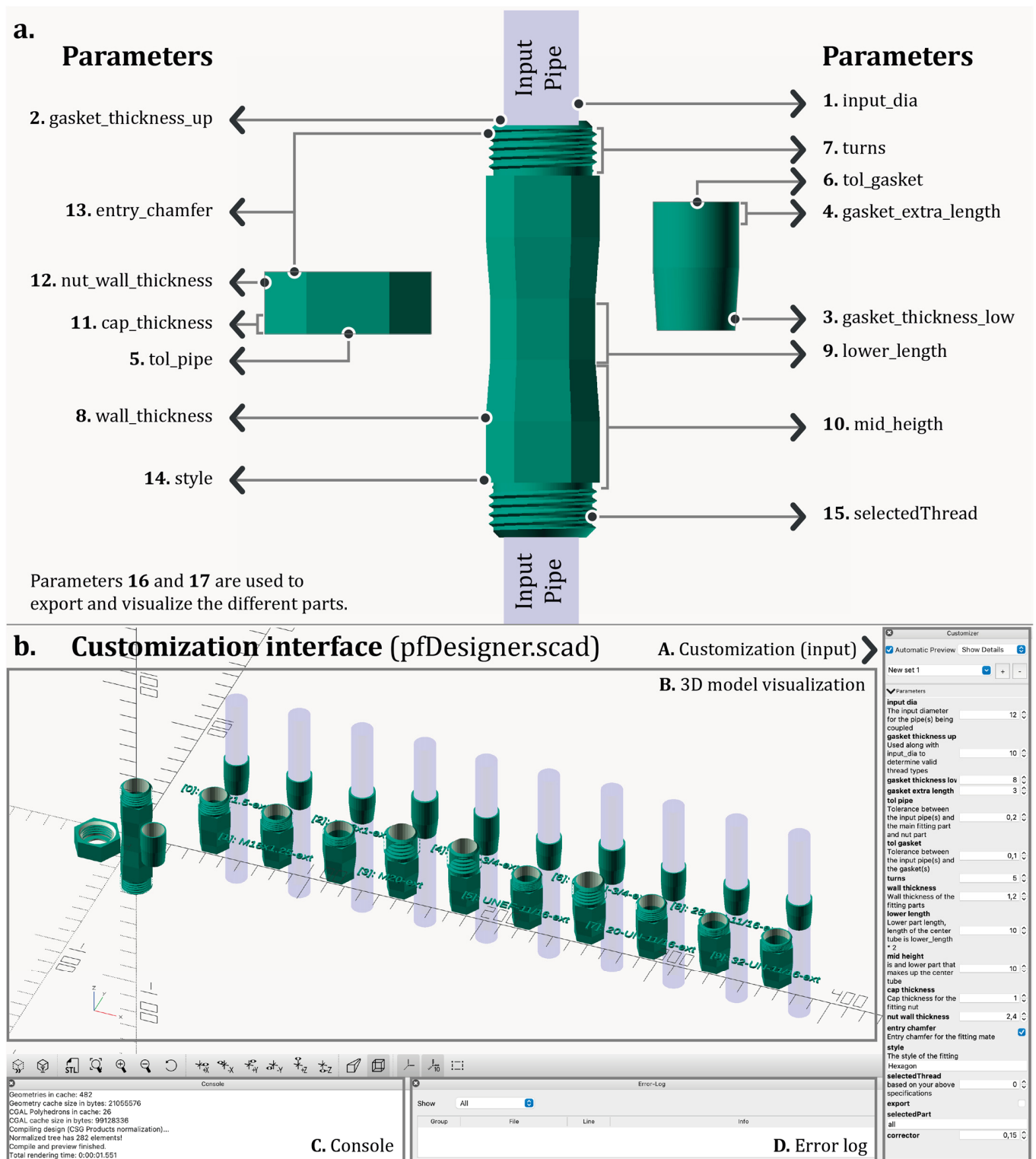


Figure 2. Customization of the pipe fitting files: (a) list of the labeled parameters of the pfDesigner.scad file modifiable by the users, showing their connection to the features of the 3D model affected by their changes (indicated with the dots and according to the classification of Table 2), and (b) visualization of the pfDesigner.scad file in the OpenSCAD environment when the user interacts with the customizable 3D model, showing the windows for the input parameters (A), the 3D model visualization with the available standard threads for the customized parts (B), the status of the computation and rendering times (C), and the error log for issue detection in the code and the parameter configuration (D).

The user, at the minimum, must enter the desired input diameter and the upper and lower range for the gasket's outer thickness, adjusting the fitting of the gasket according to the flexible material selected for its fabrication. Figure 2b shows the visualization of the parametric definition in the OpenSCAD environment. The design interface file automatically renders all valid input options for the input parameters (Figure 2b, part A), generating three 3D models for each of them in the 3D model visualization window (Figure 2b, part B): the cap nut part (Figure 2a, on the left), the fitting part (Figure 2a, in the middle), and the gasket part (Figure 2a, on the right). As a note, using a large gasket thickness range may result in a large number of valid solutions, which could increase the rendering times.

2.2. Pipe Fitting Customization and 3D Printing

Two batches of pipe fitting parts were customized and 3D printed to test the validity of the parametric definition and the reliability of the customizable design and to validate the approach. The pipe fitting 3D models were customized using the parametric design from Section 2.1. The two batches were customized to demonstrate the use of the parametric design with different kinds of pipes to be fit, which means selecting the “input_dia” parameter as the main variable. This choice led to two solutions working with different pipe diameters with a significant difference in the part dimensions, requiring different manufacturing times and material quantities. The parameters for the customization are shown in Table 3 (first half).

Table 3. Selected parameters used for the customization of the two sample batches and their main 3D printing parameters according to the specific part (pipe fitting, nut, gasket) and material (PETG, PLA, ASA, TPE).

Steps	Parameters (Input Name)	Unit	Batch 1 (10 mm Diameter)	Batch 2 (20 mm Diameter)
Customization (OpenSCAD)	Input diameter (input_dia)	mm	10	20
	Gasket thickness up (gasket_thickness_upper)	mm	17	19
	Gasket thickness low (gasket_thickness_lower)	mm	15	15
	Gasket extra length (gasket_extra_length)	mm	6	8
	Tolerance pipe (tol_pipe)	mm	0.5	0.5
	Tolerance gasket (tol_gasket)	mm	0.1	0.1
	Number of turns (turns)	//	5	5
	Wall thickness (wall_thickness)	mm	2	2
	Lower length (middle_length)	mm	12	12
	Mid height (middle_length)	mm	5	7
	Cap thickness (cap_thickness)	mm	3	4
	Nut wall thickness (nut_wall_thickness)	mm	4	6
	Entry chamfer (entry_chamfer)	//	enabled	enabled
	External cross-section style (style)	//	hexagon	hexagon
	Selected thread (selectedthread)	//	5	5

Table 3. Cont.

Steps	Parameters (Input Name)	Unit	Batch 1 (10 mm Diameter)			Batch 2 (20 mm Diameter)		
3D printing (Prusa slicer)	Materials	//	PETG	PLA	ASA	TPE	PETG	TPE
	3D printer (Prusa)	//	XL	XL	XL	MK3S	XL	MK3S
	Nozzle diameter	mm	0.4 (applied to all columns)					
	Extruder temperature	°C	240	215	260	230	240	230
	Bed temperature	°C	80	60	105	50	80	50
	Speed	mm/s	65	65	65	30	65	30
	Infill (type)	//	Rectilinear (applied to all columns)					
	Infill (overlap)	%	20 (applied to all columns)					
	Infill (percentage)	%	100 (applied to all columns)					
	Flow (percentage) ^a	%	115–110	115–110	110–100	10	110–100	100
	Layer height	mm	0.15	0.15	0.15	0.2	0.15	0.2
	Perimeters	//	6	6	6	4	6	4
	Adhesion ^a	//	Brim—none	Brim—none	Brim—none	Brim	Brim—none	Brim

^a = For PETG, PLA, and ASA, the first value refers to the pipe fitting, whereas the second one refers to the nut.

After the input selection and rendering, the 3D models were exported as STL files for slicing. The customized parts for the two batches, Batch 1 (10 mm diameter) and Batch 2 (20 mm diameter) are shown in Figure 3 and are available in the OSF directory [58]. Each rendered pipe fitting solution comprises one cap nut part, one fitting part, and one gasket part, whereas each manufactured assembly includes two cap nut parts, one fitting part, and two gasket parts. Batch 1 comprises three different sets of manufactured assemblies, which differ for the feedstock materials of the cap nut and fitting parts, i.e., PLA, PETG, or ASA. Batch 2 comprises one set of manufactured assemblies using the feedstock material with the best test performances from Batch 1, which was PETG. After preliminary tests using thermoplastic polyurethane (TPU) filaments, the gasket parts were all manufactured using the same flexible material, i.e., TPE 85 Shore A, to ensure a tight seal with the pipe fitting under pressure, good elasticity to allow its deformation without breaking, and the correct grip on the input pipes. The material was selected according to its hardness, considering the range of commercially available flexible materials and the state-of-the-art in spare part 3D printing [60–62]. PLA [63], PETG [64], and ASA [65] filaments for MEX FFF were supplied by Polymaker (Shanghai, China), whereas TPE [66] was purchased from 3D Printing Canada (Hamilton, ON, Canada). At least three copies were manufactured for each set of assemblies to ensure repeatability, resulting in nine samples for Batch 1 and three for Batch 2 to be tested.

The sample parts were sliced using the open-source PrusaSlicer 2.8.0 (Prusa Research, Prague, Czech Republic) [67]. The gcode files were 3D printed using open-source RepRap class small-format 3D printers, i.e., Prusa XL and Prusa i3 MK3S from Prusa Research equipped with a 0.4 mm nozzle. Being a Bowden extrusion system, the Prusa XL was used for the rigid filaments, whereas the direct drive extrusion MK3S was selected to manufacture TPE. The main 3D printing parameters are summarized in Table 3 (second half), and the corresponding PrusaSlicer profiles with the full list of the 3D printing parameters are available in the OSF directory [58]. The 3D printing parameters were selected to ensure good quality of the 3D-printed parts without leakages due to the MEX FFF 3D printing process, e.g., extrudate flow inconsistencies or excessive interlayer voids. The values were defined according to the specific filament datasheet, e.g., extruder and bed temperatures, and the watertight constraint to ensure 100% dense parts, e.g., speed, infill specifics, flow, layer height, and perimeters. More details about the selection process and the rationale

are discussed in Section 3.1. Information regarding the amount of material used and the 3D printing time of each component is given in Section 3.3, i.e., nominal weight, measured weight, and 3D printing time (Table 4).

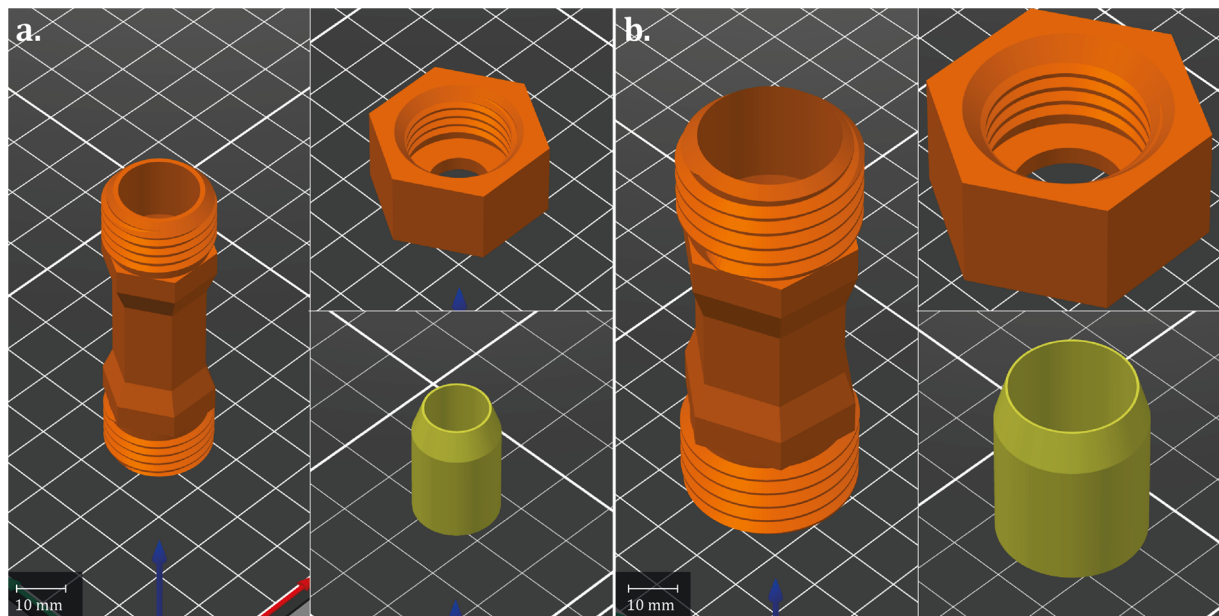


Figure 3. STL files of the customized pipe fittings obtained from the pfDesigner.scad file considering (a) a 10 mm diameter pipe and (b) a 20 mm diameter pipe (from top left to bottom right: fitting part, cap nut part, and gasket part). Size of the grid squares: 10 × 10 mm. Arrows in the figures are orientation vectors.

Table 4. Economic analysis of the pipe couplings produced from different materials (PLA, PETG, ASA, and TPE gaskets).

Custom Design	Material	Parts	Material	Weight (Nominal)	Weight (Measured)	Time	Cost ^a
Batch	Unit	//	//	g	g	min	CAD
Batch 1 (10 mm diameter)	PLA	Pipe fitting (×1 main body)	PLA	12.2	10.3	82	0.31
		Nut (×2 parts)	PLA	21.4	20.2	61	0.58
		Gasket (×2 parts)	TPE	3.9	3.2	45	0.15
		Total assembly	//	37.5	33.7	188	1.03
	PETG	Pipe fitting (×1 main body)	PETG	11.7	10.6	80	0.32
		Nut (×2 parts)	PETG	20.4	19.7	98	0.58
		Gasket (×2 parts)	TPE	3.9	3.2	45	0.15
		Total assembly	//	36.0	33.5	223	1.04
	ASA	Pipe fitting (×1 main body)	ASA	10.1	8.4	113	0.35
		Nut (×2 parts)	ASA	17.6	16.2	94	0.65
		Gasket (×2 parts)	TPE	3.9	3.2	45	0.15
		Total assembly	//	31.6	27.8	252	1.15

Table 4. Cont.

Custom Design	Material	Parts	Material	Weight (Nominal)	Weight (Measured)	Time	Cost ^a
Batch 2 (20 mm diameter)	PETG	Pipe fitting (×1 main body)	PETG	28.3	29.0	146	0.85
		Nut (×2 parts)	PETG	69.8	71.2	296	2.07
		Gasket (×2 parts)	TPE	11.2	8.4	125	0.38
		Total assembly	//	109.3	108.6	567	3.30

^a = for a detailed explanation of the cost calculation, please see Section 2.4.

2.3. Pipe Fitting Testing Experimental

The 3D-printed pipe fittings underwent hydrostatic water pressure tests to validate their use for residential, agricultural, and renewable energy production applications. Due to the considerable amount of energy stored in a pressurized air tank, it is prohibited by most international manufacturers and standards to test any plastic pipeline using pressurized air [68,69]. As is shown in Figure 4, all the hydrostatic pressure tests were conducted under water pressure using a VEVOR hydraulic pressure test hand pump (Shanghai, China) capable of increasing the water pressure up to 5 MPa. Figure 4b shows the control system of the test equipment, which consists of a calibrated 3-unit pressure gauge as well as a twin-valve system that enhances reading accuracy and helps with leakage inspection by trapping water and maintaining hydrostatic pressure. The two main batches of 3D-printed pipe fittings underwent at least 16 pressure test processes to ensure the reliability of the test results and the reusability of the 3D-printed fittings and TPE gaskets, comparing the different materials and pipe sizes.



Figure 4. Pressure test: (a) VEVOR hydrostatic bench test setup with calibrated pressure gauge and maximum pressure range of 5 MPa; (b) control system of the equipment.

The pressure test was conducted using the same testing conditions for all 3D-printed fittings sizes and materials. It started with a deaeration process by installing the pipe fitting vertically and loosening the top-end nuts until any trapped air inside the test equipment was removed when water was being pumped into the pipes. Accordingly, the main pressure test was conducted in multiple steps for each coupling so that the pressure increased by 0.345 MPa at each step until any leakage or pressure drop in the equipment was detected. The test was repeated at different torque values, ranging from 5 to 15 Nm for Batch 1 and from 7.5 to 17.5 for Batch 2. Also, the time span between the steps was 180 s to ensure that the fitting could maintain the pressure without noticeable leakage. The pre-installed

pressure gauge on the hand pump had an accuracy of ± 20 psi and was calibrated before conducting the experiments.

2.4. Economic Analysis Method

An economic analysis was performed on the two batches of 3D-printed pipe fittings to validate their suitability in real-world distributed manufacturing contexts, such as developing countries and remote areas, following the method described in [70], which includes material and fabrication electricity costs. The 3D printing times and costs were quantified for each set of samples from both batches, starting from the nominal times and weights reported from the slicing in PrusaSlicer. The nominal times were compared to the real ones, keeping the higher values for the analysis. The nominal weights were calculated using the densities of PLA, PETG, ASA, and TPE provided in the datasheet, which were 1.31 g/cm^3 , 1.25 g/cm^3 , 1.15 g/cm^3 , and 1.15 g/cm^3 , respectively. The actual values were also obtained by weighing the 3D-printed parts to acknowledge differences between nominal and real values. The material cost for the economic analysis was considered to be CAD 27.99 for PLA and PETG [63,64], CAD 38.99 for ASA [65], and CAD 41.98 for TPE per kg of filament [66]. The power consumption of the printers was considered to be 120 W according to the nominal power consumption specified by the manufacturer [71]. Finally, the cost of electricity was extracted from the datasheet provided by the Ontario Energy Board from the mid-peak price period, which is equal to CAD 0.122/kWh [72]. The weight and time data used for the economic analysis are reported in Section 3.3 (Table 4).

3. Results and Discussion

3.1. Open-Source Parametric Pipe Fittings

Figure 5 shows the 3D-printed parts for both batches obtained from the parametric design in OpenSCAD. The different components were fabricated without the need for support, optimizing the material usage, minimizing waste, and reducing postprocessing timings. As shown in Figure 5a–c, the parts can be 3D printed with different feedstock materials, ranging from more common and inexpensive filaments, e.g., PLA or PETG, to high-performing ones, such as ASA. In addition, the flexible gaskets (Figure 5d) were successfully 3D printed using a medium flexible TPE filament (85 Shore A hardness), and bigger parts can be manufactured without visible defects or the need for supports (Figure 5e). These design choices help make parametric design more accessible to entry-level users and reduce the time required to produce customized parts. Furthermore, avoiding supports can contribute to obtaining smoother surfaces and reduce the number of defects due to material removal, increasing the pipe fitting lifespans.

Fluid leakages must be prevented to ensure the reliability of the pipe fitting design during use. To this end, different manufacturing strategies were combined to reduce the risk of leakages of the single parts and the assembled components, as detailed in Table 3. First, the rigid parts were 3D printed with 100% rectilinear infill, aiming to achieve solid parts without internal voids. The extrusion flow, or flow multiplier, was also increased by 5–15% to reduce the risk of interlayer porosity, detachment, or delamination between the different layers and extrusion toolpaths, in line with the state-of-the-art focused on MEX FFF process optimization [73]. The overlap between the infill and the perimeters was then adjusted to 20% to achieve a homogeneous deposition within the different cross-sections of the parts. These choices are in line with MEX FFF, as the extrudate bonding between the layers and extrusion toolpaths can be affected by inconsistencies, localized defects, or shrinkage, which may lead to interlayer porosity [73]. In addition, fine-tuning 3D model tolerances facilitated a tight connection between the different parts, especially considering the influence of an increased extrusion flow multiplier on the actual geometry of the

components [73]. The layer height was selected to ensure good resolution of the threaded parts in the z-axis direction and, at the same time, limit the interfaces between the layers, reducing the bonding parts and, hence, the potential interlayer voids [74]. Finally, the TPE gaskets further reduced the assembly leakages, allowing their deformation under pressure to ensure a tight seal between the pipe fitting and nut, thanks to the specific material and thickness of the gasket. As mentioned in Section 2.2, the material was selected according to its Shore A hardness, ensuring a watertight seal of the fitting part, the correct grip on the input pipes, and good elasticity for its deformation under pressure without failures. This choice is supported either by the current market options, which offer commercial TPE filaments with different hardnesses, and the feasibility of printing customized spare parts made of flexible materials, including gaskets for research and lab environments [60–62]. For these reasons, the parts did not undergo further postprocessing, such as coatings or heat treatment [75]. This aspect makes the parts easier to manufacture in distributed contexts, requiring low-cost MEX FFF 3D printers and fewer raw materials.

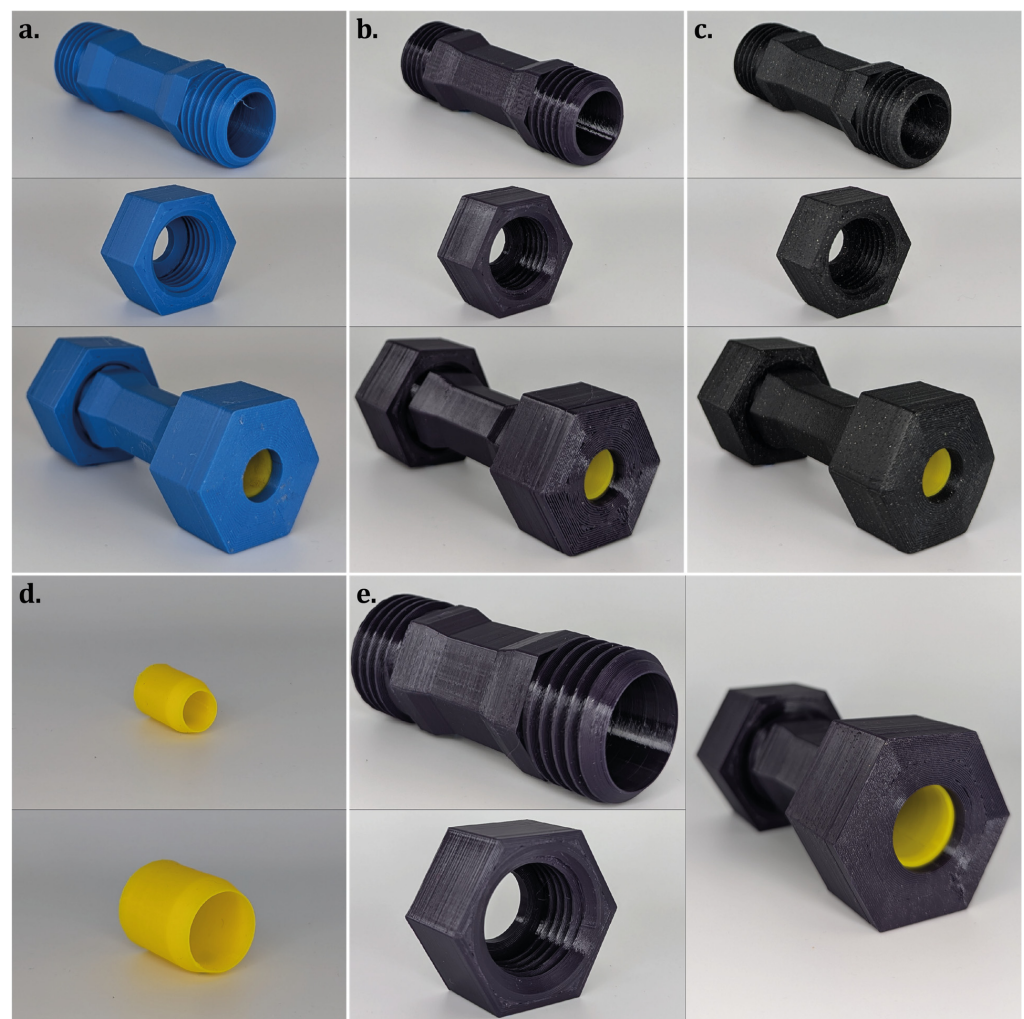


Figure 5. 3D-printed pipe fitting parts for the test: (a) 10 mm-diameter pipe fitting parts made with PLA (from top to bottom: pipe fitting, nut, and assembled parts with the TPE gasket); (b) 10 mm-diameter pipe fitting parts made with PETG (from top to bottom: pipe fitting, nut, and assembled parts with the TPE gasket); (c) 10 mm-diameter pipe fittings made with ASA (from top to bottom: pipe fitting, nut, and assembled parts with the TPE gasket); (d) 10 mm (top) and 20 mm (bottom) gaskets made with TPE; and (e) 20 mm-diameter fitting parts made with PETG (from top to bottom: pipe fitting, nut, and assembled parts with the TPE gasket).

3.2. Pipe Fitting Testing

During the pressure test process, the effect of material and pipe dimension was investigated on the highest pressure tolerated by the 3D printed fittings. To check the influence of material on the maximum pressure tolerated by fittings, the parts were 3D printed using PLA, PETG, and ASA based on a pipe diameter of 10 mm and tested under the same working conditions (Batch 1). The fittings were tightened using a torque wrench to investigate the relationship between the tightening torque and the maximum pressure before any leakage, and the torque was raised until physical damage, such as cracking, was observed. The pressure test results for the 10 mm pipe fittings in Figure 6a show that the PETG offers the best performance by tolerating up to 4.551 ± 0.138 MPa of water pressure, followed by ASA and PLA with 2.758 and 2.275 MPa, respectively. The figure also shows that the ASA can be tightened to up to 15 Nm of torque, while the highest torque for PETG and PLA parts was 12.5 and 7.5 Nm, respectively, before any cracks appeared (refer to Figure 6b–d for the corresponding broken parts). To check the effects of pipe size on the pressure tolerance of the fitting, another sample part was 3D printed using PETG based on a 20 mm pipe diameter (Batch 2). Figure 7a shows that the highest pressure achieved by this coupling was as high as 2.392 ± 0.138 MPa under 17.5 Nm of torque. The lower pressure achieved by 20 mm coupling (Batch 2) compared to the smaller version (Batch 1) can be explained by the fact that larger gaskets need a larger amount of torque to seal the coupling, which explains the cracking in the body of the 3D-printed coupling in Figure 7b.

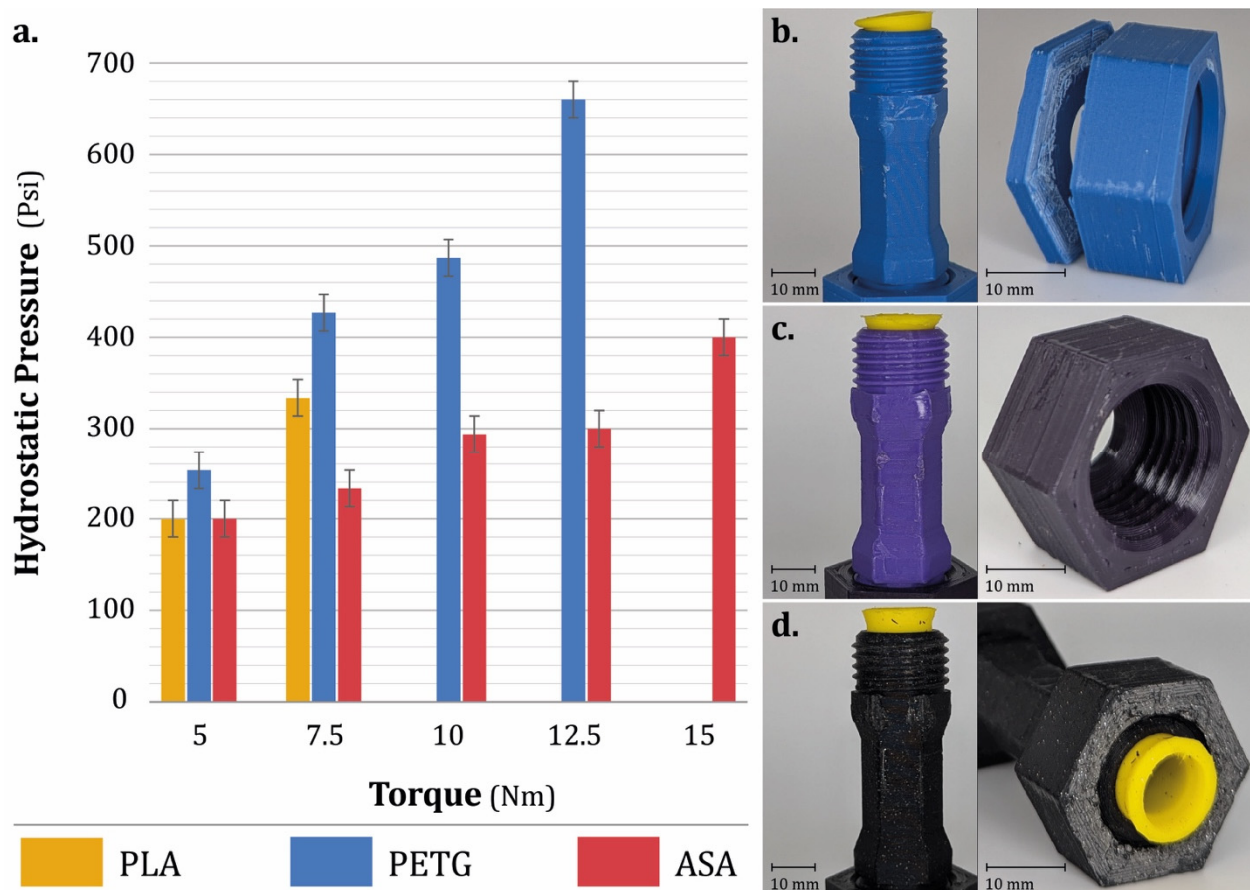


Figure 6. Hydrostatic pressure tests on Batch 1 (10 mm diameter): (a) average results over time for PLA/TPE, PETG/TPE, and ASA/TPE pipe couplings at different torques; 3D-printed parts after the test, showing the status of the components or failures (main body on the left and nut on the right) of (b) fitting parts made with PLA/TPE, (c) PETG/TPE; and (d) ASA/TPE.

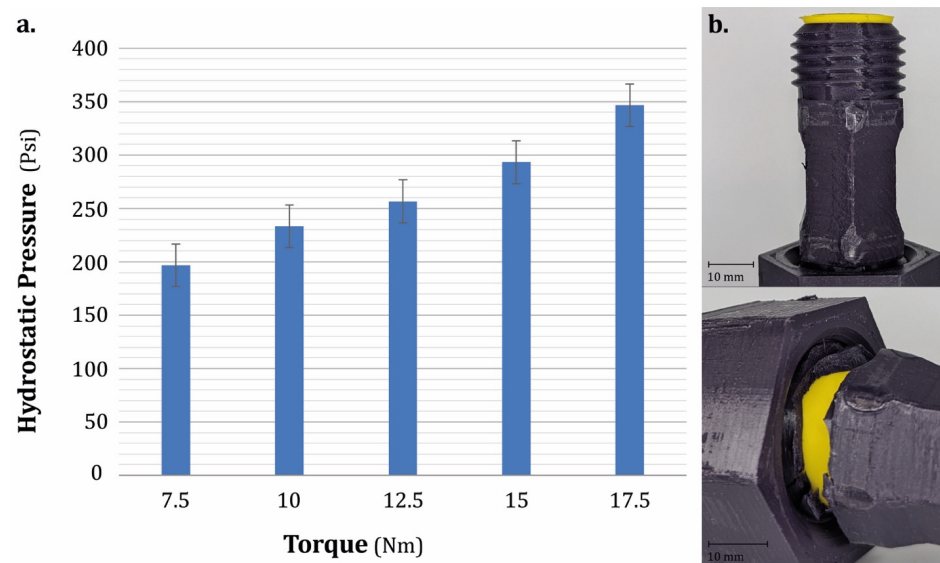


Figure 7. Hydrostatic pressure tests on Batch 2 (20 mm diameter): (a) average results over time for PETG/TPE pipe couplings at different torques; 3D-printed parts after the test, showing the status of the components or failures (main body on the right and nut on the left) of (b) fitting parts made with PETG/TPE.

3.3. Economic Analysis

The proposed design and manufacturing method for the pipe fittings offers both direct and indirect cost benefits. To show the direct economic benefit, the overall cost of a pipe fitting was calculated and is presented in Table 4, which includes the weight of the couplings, the material cost, the printing time, and the cost of electricity in Ontario, Canada. As is shown in Table 4, the cost of a complete 3D-printed pipe fitting for the outer pipe diameter of 10 mm varies from CAD 1.03 for PLA to CAD 1.15 for ASA. A comparison between the cost of a 3D-printed pipe fitting and similar commercial pipe couplings (presented in Table 5) clearly shows that the price of a commercial pipe fitting for a similar pipe size and a much smaller pressure range can be between four and seventeen times higher than that of a 3D-printed part. Furthermore, the ease of 3D printing a pipe fitting in remote areas where shipping services are either unavailable or expensive, as well as the fact that installing the printed parts does not require any special tools or level of expertise, emphasizes the indirect economic benefits of the proposed approach.

Table 5. List of commercially available water pipe couplings for price comparison and economic analysis purposes.

Specifications	Vendor	Material	Pipe Diameter	Pressure Rating [MPa]	Price [CAD]
Rain Bird Easy Fit Compression Fitting [11]	RONA	ABS	16–17 mm	0.414	4.79
John Guest Push-to-Connect Union Fitting [13]	Home Depot	Polypropylene	1/4 inch	1.034	5.13
SharkBite ProLock Push-to-Connect Plastic Coupling [14]	Home Depot	Polymer	1/2 inch	1.103	13.16
SharkBite Max Brass Push Coupling [15]	Home Depot	Brass	1/2 inch	1.724	13.68
IPEX Philmac Quick Disconnect Compression Coupling [16,76]	RONA	polypropylene	1/2 inch	1.586	17.49

Table 5. *Cont.*

Specifications	Vendor	Material	Pipe Diameter	Pressure Rating [MPa]	Price [CAD]
Aqua-Dynamic 3/4" Brass Push-Fit Coupling [17]	KENT	Brass	1/2 inch	1.379	17.99
SharkBite EvoPEX Connector Push-to-Connect [12]	Amazon.ca	Stainless Steel	1/2 inch	1.379	18.44

3.4. Impact, Application Context, and Feasibility

According to the International Residential Code (IRC), static water pressure should not be greater than 0.552 MPa. If it exceeds this limit, a pressure-reducing valve should be installed to lower the static water pressure inside buildings [77]. Similarly, because the flow rate of the working fluid inside the photovoltaic thermal (PVT) systems is considerably lower than that of the residential sector, the hydrostatic fluid pressure in these systems usually varies from 0.097 to 0.303 MPa [78,79]. In addition, most irrigation systems inside greenhouses, such as drip irrigation systems, operate best between 0.138 and 0.207 MPa [80]. From this perspective, the new open-source pipe fitting design introduced in the current study can withstand more than 8.2 times the maximum standard pressure in a building and considerably higher pressure than any greenhouse or PVT system requires. Due to the flexibility of the parametric design and the compatibility of the current design with different materials and pipe diameters, the introduced fittings could be the fastest and least expensive option to build a liquid supply system or to repair a leakage. As a real-world application, Figure 8 shows a pipe fitting installed on the inlet water line of a waterjet cutter device, connecting two pipes with different materials and slightly different outer diameters to each other under approximately 0.414 MPa of tap water pressure. It should be pointed out that all experiments were conducted using a pH of seven. In some regions, water has different pH values either due to the supply or to prevent corrosion [81]. Previous research on the chemical compatibility of 3D printing plastics has shown that the materials used in this study would be able to withstand much more severe chemicals (e.g., solutions of over 30% HCl) [82]. To use these couplings for chemical applications or applications with extremely high pressures, future work may be necessary.

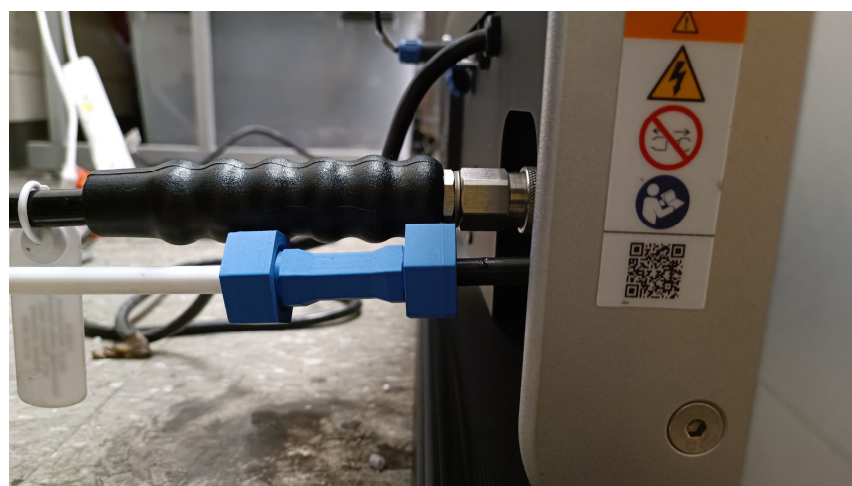


Figure 8. Practical application of the couplings by connecting two plastic pipes of different materials to a waterjet cutter as a water inlet line.

3.5. Future Work

Although further optimization should be performed in the future, this study introduces the concept of distributed customizable 3D-printed pipe fittings for water manage-

ment system applications. The open-source CAD design enables the manufacture of other types of pipe fittings by using minor code modifications, e.g., three-ways or T junctions. It also allows the user to connect pipes with different outer diameters using only one pipe fitting, which is much more compact and less expensive than commercial pipe couplings.

Apart from the technical and economic advantages of the new design, its independence from equipment provided by distributed manufacturing methods is another plus point for the current approach. Enabling anyone to design and produce equipment tailored to an existing water supply system makes the current approach capable of being used in space exploration applications, as well as in remote locations, without requiring access to standard commercial equipment. In addition, thanks to the current user-friendly design, people with any level of technical skill can design or repair their own water supply systems without paying excessive amounts of money for installation and maintenance services.

The environmental impacts of the current design would be considerably lower than those of commercial plumbing equipment available on the market. For instance, the current pipe coupling can also be 3D printed using recycled plastic [49,51], and all the equipment used to turn the waste plastic into filaments, as well as the 3D printers, can be powered by renewable electricity, e.g., photovoltaics [46,50]. Despite the need for further tests, the 3D-printed pipe fittings can also be mechanically recycled at the end of their life, resulting in new secondary raw materials for MEX 3D printing, e.g., flakes, pellets, or filaments [49]. The resulting recycled materials could then go through multiple recycling processes to keep these resources in use through time [83–85], reducing the need for virgin materials for this specific application. These choices can further reduce material feedstock costs, paving the way for the optimization of this application using secondary raw materials from commodity waste. It should be noted, however, that toxicity and environmental safety should be considered for recycled plastics [86,87] and composites [88], providing a rich area of future work to create a circular economy [89].

Although the current design offers higher pressure tolerance compared to what is required for buildings, greenhouses, and renewable energy production applications, an optimization analysis of the fitting mechanism and the dimensions of the pipe coupling can further reduce the production costs as well as the carbon footprint of the new design compared to conventional approaches.

4. Conclusions

This study introduced an open-source parametric design for 3D-printed pipe fittings composed of three components: a coupling, nuts, and a gasket. These components were made of different thermoplastic materials, namely, PETG, PLA, ASA, and TPE, compatible with waterpipes of various materials and outer diameters. All the printed parts passed numerous hydrostatic pressure tests, and the test results, as well as the economic analysis of these parts, resulted in the following conclusions:

- The parametric design adds substantial flexibility to the piping system while adhering to the existing standards for interoperability.
- The pressure test results show that PETG couplings can tolerate up to 4.551 MPa of hydrostatic pressure, which is eight times greater than the highest standard water pressure for the residential sector, followed by ASA and PLA that endured 2.758 and 2.275 MPa, respectively, without exhibiting any leakage.
- Based on the economic analysis, the cost of 3D printing a pipe coupling for a pipe diameter of 10mm ranges from CAD 1.03 for PLA to CAD 1.15 for ASA, which is up to 17 times lower than purchasing a commercially available pipe fitting of a similar size.

- The independence of the commercial supply chain in developing countries and the ability to mitigate the drawbacks of distributed manufacturing methods in agricultural industries in rural habitats or space applications can be seen as a social and indirect economic benefit of the current study.
- Future work could use waste plastic to 3D print pipe fittings at even lower costs while producing piping and plumbing components with a considerably lower carbon footprint.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

OpenSCAD, which is a free and open-source software package, operates by providing the specification of geometry using code in a script. OpenSCAD utilizes its own custom scripting language, which is designed for functional programming, parametric modularization, and mathematical computation. This language, based on a declarative **syntax**, allows users to define variables, functions, and modules using constructs such as *var = value*; *function myFunction(parameters) = myExpression*; and *module myModule(parameters) { myActions }* to create complex, reusable models. Variables in OpenSCAD are scoped, and within each scope, they are immutable; in other words, variable names can be reused if properly contained within their scope, and once assigned, a variable’s value cannot be changed. OpenSCAD supports modularity and code reuse with *include <myfile.scad>*, which imports and executes all code from an external file, exposing variables, functions, and modules, while *use <myfile.scad>* loads a file but only makes its modules and functions available, preventing unintended execution. **Operators** such as $n + m$ for addition and $n < m$ for less than comparisons can be used to perform mathematical operations. Additionally, **Special variables** like *\$fa* for minimum angle and *\$fs* for minimum size can be used to control the rendering of models. OpenSCAD supports both **2D** and **3D** modeling, with functions like *circle(radius | d = diameter)* and *sphere(radius | d = diameter)*, allowing users to create complex shapes. **Transformations** like *translate([x,y,z])* and *rotate([x,y,z])* can be used to manipulate objects in 3D space. The language also includes **Lists**, which can be created and indexed using *list = [...]*; and *var = list [2]*; and **Boolean operations** like *union()* and *difference()* can be used to combine and subtract objects. Furthermore, OpenSCAD includes **Flow Control** statements like *for (i = [start:end]) { ... }* and *if (...) { ... }*, as well as **Type test functions** like *is_undef* and *is_bool*, and **Mathematical functions** like *abs* and *sin*, which can be used to create complex and robust models.

Figure A1 illustrates the workflow for creating a parametric model in OpenSCAD using the pipe fitting file as an example. It begins with the parameterization of the objects (Step A), where the desired geometry is sketched out and dimensions are assigned variables in a .scad definition (Step B). These parameters are then written in the OpenSCAD editor (Step C), and these parameters are used to construct the geometry (Step D). In the presented example in Figure A1, the base_shape module accepts these parameters (Step C) to construct the 3D shape based on 2D polygons, defining the geometry of the objects with points calculated based on the input values, which is then transformed into a 3D model. The output is displayed to the OpenSCAD interface as a preview of the resulting 3D mesh, which can be rendered and exported as a mesh for external use or fabrication (Step E).

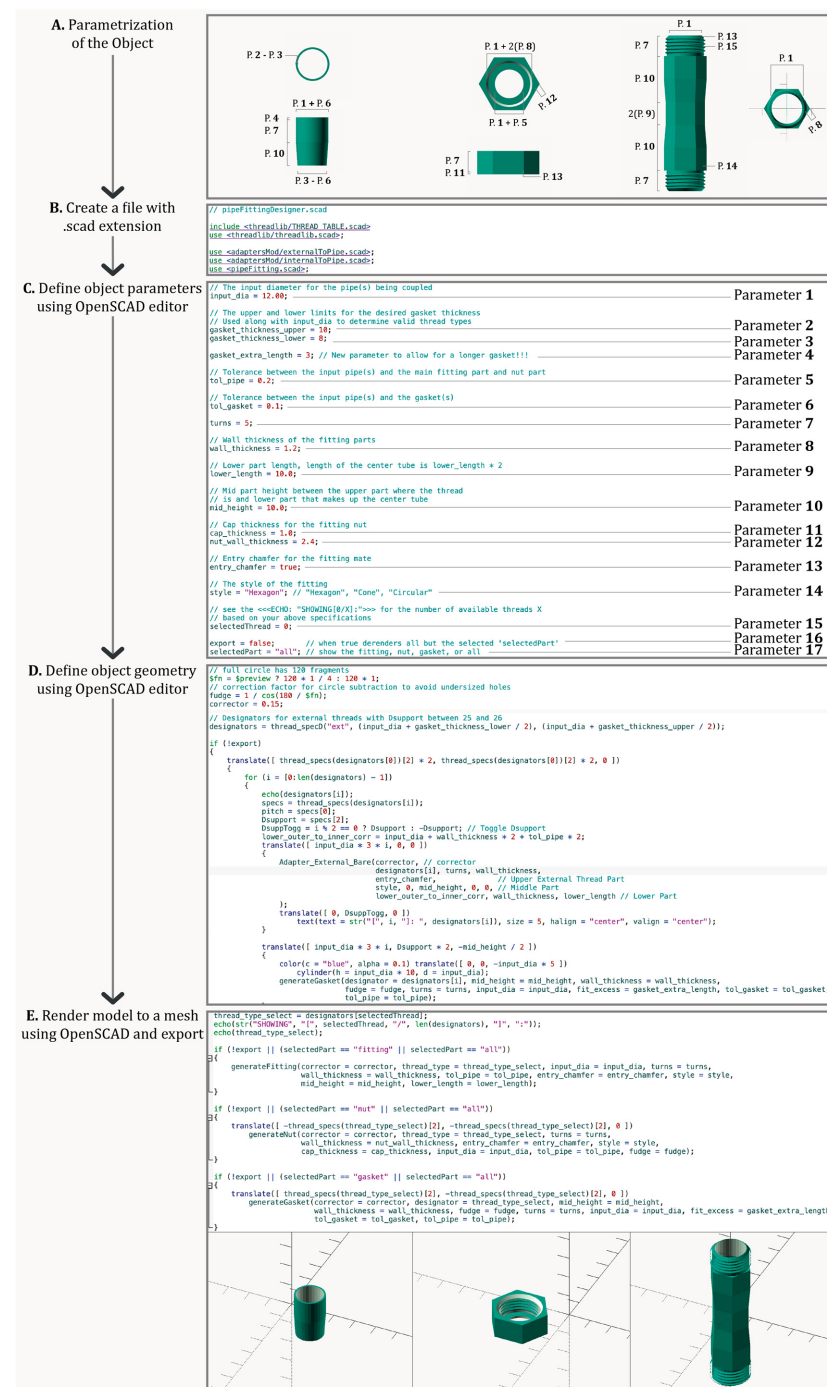


Figure A1. Workflow for parametric design of the pipefitting 3D models in OpenSCAD.

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